I started by casting a wide net rather than assuming upfront which stats would matter, pulling in season-level pitch-tracking data on both pitchers and hitters, minor-league performance data, and pitch-by-pitch outcome counts, then let the results show what was actually worth keeping. From there I built each predictive variable from a pitcher's past seasons only, never the season being predicted, since the whole point is forecasting 2025 from what was already known before it started. That includes a pitcher's own K%, walk rate (BB%), and several swing-and-miss stats (whiff%, zone-swing-miss%, and other plate-discipline rates) from the prior season and the season before that, along with the average velocity and spin of each pitch type in his arsenal, so a change in stuff shows up before it shows up in results. Total batters faced (TBF), both that season and career-to-date, is included directly since a pitcher's role and consistency tend to firm up with more experience. All of this is compared against the league-wide strikeout rate from the season before, so the model always has a sense of what "normal" looked like at the time.

For the prediction method itself, I combined three techniques, each good at catching something different. Empirical Bayes shrinkage pulls a pitcher's own recent K% toward the league average, protecting against overreacting to a small sample. Elastic net regression automatically zeroes out variables that don't add real information. Gradient-boosted trees (gbm) pick up more complicated, non-linear patterns and interactions. The three are blended with non-negative weights chosen to minimize error. To test honestly, I used a rolling-origin backtest, training on seasons through 2022 and validating on 2023, then training through 2023 and validating on 2024, simulating how the model would have been used in real time. gbm's hyperparameters (tree depth, learning rate) were tuned inside that same backtest rather than gbm's default random cross-validation, and 2025 was opened only once, at the very end, to score the finished model.

Running that first version against a mixed population of pitchers exposed a problem, since a single model treated an established starter, a true rookie with no MLB history, and a position player making a one-off pitching appearance the same way, and it showed in the accuracy. So I split pitchers into five groups by how much history they had: an established pitcher with a full model, a true rookie with no history, and three variants for position players, minor-league-only pitchers, and pitchers with no usable record, then built out what each group needed. For position players and minor-league pitchers, that meant matching in outside data, a separate hitting leaderboard matched by player to identify position players by career plate appearances, and a minor-league dataset matched the same way to give minor-league-only pitchers a translated projection instead of leaving them with no history. I also went back and added one more feature to the main model at this stage, CSW% (called-plus-swinging-strike rate), the share of all pitches thrown that result in a called or swinging strike, a strong predictor of future strikeouts that doesn't come pre-built anywhere. I built it myself from raw pitch-outcome counts (balls, called strikes, swinging strikes, fouls) and matched it in by player ID and season. My first attempt matched pitchers by normalized name instead, which meant manually resolving about ten same-name collisions in a season. Once I found the same counts available under a stable player ID already used elsewhere in the data, I switched to that cleaner join.

With the five groups in place, I kept iterating on the blend. I tested a TBF-adaptive version that lets the weights on the three techniques shift by batters-faced tercile instead of using one fixed set, and kept it only because it beat the fixed global blend on the same backtest. I also tested a per-bucket bias correction, adjusting each group's predictions by its own mean signed backtest error from 2022-2024. It improved

accuracy for every group except confirmed position players, where the correction was measured on a differently defined historical population and made 2025 predictions worse, so I left it off for that group rather than applying it everywhere by default.

On the 2025 season, held out completely and never used to build the model, predictions missed by about 4.2 percentage points on a typical pitcher (RMSE 0.0416), with a typical individual miss of about 3.2 percentage points (MAE 0.0321), for example predicting 22% for a pitcher who actually struck out 26.2% of batters. The model explained about 40% of the differences in strikeout rate from one pitcher to the next (R^2 0.40), a meaningful amount for a stat this volatile, though it leaves real room for a pitcher to outperform or underperform what history alone suggests. What drove the predictions was mostly a pitcher being himself. A TBF-weighted trailing K% was by far the top gbm driver (37% relative influence), followed by last season's raw K% (17%) and last season's StuffPlus, a pitch-quality score built from velocity, spin, and shape (10%). Whiff% and zone-swing-miss% each added a real amount, and fastball velocity mattered more than any other single arsenal metric. CSW%, despite being a legitimate predictor of strikeout ability on its own, added under 1% relative influence in gbm and was zeroed out by the elastic net, largely redundant with whiff% and StuffPlus once those are already in the model. By group, established pitchers with a full MLB track record (671 pitchers) were predicted most accurately. Confirmed position players (55) and minor-league-translated pitchers (86) were noisier, as expected given thinner histories. The group with no usable record (59) was hardest to predict, which follows directly from having the least information to work with. A separate statistical check also shows a real chunk of the remaining error isn't fixable by a better model, since pitchers who face very few batters carry much more pure random noise in their season K% than pitchers with a full workload.

If rebuilding with the same structure, I'd replace the roughest part of the current approach (translating limited minor-league season totals into an MLB-equivalent projection) with more detailed pitch-by-pitch minor-league data where it exists, add context about the quality of hitters a pitcher actually faced (a tough or weak schedule isn't reflected anywhere), and bring in real injury and missed-time records, especially arm-related ones, which the model currently only sees indirectly through reduced playing time. A partial season lost to injury and a mid-season minor-league demotion look identical to it today, even though a pitcher returning from a significant arm injury tends to follow a fairly well-documented performance pattern that a dedicated flag could capture. I'd also revisit the position-player correction once another season of test data is available, since it's the one place a fix that helped everywhere else measurably hurt. Those are the pieces that target where the model is weakest today, without changing the core rule that keeps the whole approach honest. Nothing from the season being predicted is ever used to predict it.